

EDUCATION

University of Washington

PhD

Seattle, WA, USA

Sep. 2023–Jun. 2028 (expected)

University of Toronto

Honours Bachelor of Science

Toronto, ON, Canada

Jun. 2023

- Majors in computer science and mathematics, minor in statistics
- Cumulative GPA (cGPA): 3.99/4.0

PUBLICATION

- [1] **Dedong Xie**, Theano Stavrinou, Jonggyu Park, Simon Peter, Baris Kasikci, and Thomas E Anderson. “PASS: A Power Adaptive Storage Server”. *Proceedings of the 21st European Conference on Computer Systems*. 2026, pp. 1365–1380.
- [2] **Dedong Xie**, Theano Stavrinou, Kan Zhu, Simon Peter, Baris Kasikci, and Thomas Anderson. “Can Storage Devices be Power Adaptive?” *Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems*. HotStorage ’24. Santa Clara, CA, USA: Association for Computing Machinery, 2024, pp. 47–54. DOI: 10.1145/3655038.3665945.
- [3] **Dedong Xie**, Zhen Jia, Zili Zhang, and Xin Jin. “Optimizing Half Precision Winograd Convolution on ARM Many-Core Processors”. *Proceedings of the 13th ACM SIGOPS Asia-Pacific Workshop on Systems*. APSys ’22. Virtual Event, Singapore: Association for Computing Machinery, 2022, pp. 53–60. DOI: 10.1145/3546591.3547529.
- [4] Kan Zhu, Yufei Gao, Yilong Zhao, Liangyu Zhao, Gefei Zuo, Yile Gu, **Dedong Xie**, Zihao Ye, Keisuke Kamahori, Chien-Yu Lin, et al. “NanoFlow: Towards Optimal Large Language Model Serving Throughput”. *19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25)*. 2025, pp. 749–765.
- [5] Yigong Hu, You-Liang Huang, Haodong Zheng, Yicheng Liu, **Dedong Xie**, and Baris Kasikci. “Diagnosing Performance Issues in {Application-Defined} Resources”. *20th USENIX Symposium on Operating Systems Design and Implementation (OSDI 26)*. 2026, pp. 647–664.
- [6] Yi Pan, Wenbo Qian, **Dedong Xie**, Ruiyan Hu, Yigong Hu, and Baris Kasikci. “Magnetron: Optimizing Energy Efficiency of ML Systems via Differential Energy Debugging”. *24th USENIX Symposium on Networked Systems Design and Implementation (NSDI 27)*. to appear. 2027.

RESEARCH EXPERIENCE

Research Assistant

Sep. 2023–Present

Syslab & EfesLab.

Paul G. Allen School of Computer Science and Engineering, University of Washington.

Advisors: Prof. Simon Peter and Prof. Baris Kasikci

- **Research project 1:** power adaptive storage system.
- Power measurement study of power consumption patterns of datacenter storage devices: HDDs and SSDs.

- Published a first-author paper in HotStorage 2024.
- Designed and implemented a power adaptive storage all-flash server for disaggregated storage.
- Results showed that the system achieved magnitudes better performance compared with best performing baselines under power caps.
- To appear as a first-author paper in EuroSys 2026.
- **Research Project 2:** storage backbone for throughput oriented LLM serving framework
- Designed KV-cache offload to disk for NanoFlow, a throughput-oriented LLM serving framework.
- As part of the NanoFlow paper published in OSDI '25.
- **Research Project 3:** profiler for software energy waste in ML applications
- Designed a energy profiler that can trace ML application energy consumption on kernel/operator level and find root cause of software-induced energy inefficiencies.
- Leveraged CUDA graph and Pytorch Dynamo graphs for kernel replay and power measurement. Integrated traces from Pytorch profiler, CUPTI, and instantaneous power readings for analysis.
- Applied energy profiler to 9 different ML framework and applications. e.g. Pytorch, TensorFlow, vLLM, SgLang, HuggingFace Transformers, HuggingFace Diffusers.
- Accepted by NSDI '27.

Student Researcher

Jan. 2026–Jun. 2026

Google

Internship Hosts: Yong Xia and Juncheng Gu

- Designed and built a failure-resilient C++ multi-slice LLM training framework to enable flexible training and experimentation on CPU emulated hardware accelerators (e.g., GPU and TPU).
- The framework allows training cluster to continue training progress after accelerator failures without the need for instant replacement or repair.
- The framework supports parallel schemes of tensor parallelism and context parallelism.
- Contributed 20,000 lines of code additions and 40,000 total line changes across the runtime infrastructure.
- Verified system compute characteristics, ensuring throughput loss scales strictly proportional to remaining healthy devices during failure scenarios.
- Verified CPU framework numerical correctness against PyTorch baselines, achieving relative difference divergence $< 10^{-4}$ after 1000 training steps for 30 or 32 parallel devices.

Research Intern

Jul. 2022–Feb. 2023

RISELab. University of California, Berkeley.

Supervisor: Prof. Ion Stoica

- Migrated BigScience BLOOM Large Language Model to use the parallel interface provided by Alpa, a LLM training and serving framework.
- Contributed to Alpa's compliance with the latest version of transformers package and documentation of a guide on serving Alpa on Slurm clusters.

Research Assistant

May. 2022–Dec. 2022

SysNet Lab. Department of Computer Science. University of Toronto.

Supervisor: Prof. Eyal de Lara

- Participated in IBM CAS Canada project 1153 - Reducing JVM memory costs in the cloud
<https://www-40.ibm.com/ibm/cas/canada/projects?projectId=1153>
- Sole developer of the run-time memory profiler of OpenJ9 JVM JIT-compiler.
- Implemented the memory allocation logger, post-process pipeline, and visualizer with 3,000 lines of code in C++ and Python.
- Found external fragmentation and late release of memory to be main causes of memory inefficiencies.

- [Video](#) of my presentation, and [slides](#) of the presentation.

Research Intern

Jun. 2021–Jul. 2022

AI Lab. Amazon Web Services (AWS).

Supervisors: Dr. Zhen Jia (AWS) and Prof. Xin Jin (Peking University)

- Sole developer of HAWC, a half-precision Winograd convolution system for Amazon Graviton-2 ARM architecture chips.
- Proposed customized memory layout for Amazon Graviton-2 chips, ARM-specific matrix multiplication kernel generator, and minimal multi-threading scheduler to accelerate Winograd convolution.
- Implemented 3000 lines of code using C++ and ARM assembly.
- Achieved on average $11\times$ and up to $28\times$ speedup with generated matrix multiplication kernels exploited up to 89% of theoretical maximum TFLOPS of the hardware.
- Published a first-author paper in ACM Asia-Pacific Workshop on Systems (APSys 2022).
- [Video](#) and [slides](#) of my presentation in APSys 2022.

COURSEWORK AND PROJECTS

• CSE599K Systems for ML

Theme: Modern large language model (LLM) training and serving systems

Topics: Transformers, GPU basics, Serving frameworks (vLLM, DistServe, etc.), Training frameworks (ZeRO), optimization techniques (parallelism paradigms, quantization, kernel fusion), MoE and RAG

Course assignments: Implement attention with KV cache in pyTorch, profile prefill and decode attention performance on AMD MI200 GPU, RAG pipeline with Gmail data

• CSE582 Course project

Course project: Life-cycle-assessment of LLM carbon footprint

Project description: Analyzed checkpointing and different parallelization paradigms (data parallelism, tensor parallelism, pipeline parallelism) in training and serving and quantitatively calculated their usage of resources including compute, memory, network communication and disk I/O. Then attribute the utilization of computer components like GPU, CPU, NIC, SSD to model for operational energy cost and thus operational and embedded carbon.

AWARDS AND ACHIEVEMENTS

- **Faithful Steward Endowed Presidential Fellowship in Computer Science & Engineering** Sept. 2025
– Paul G. Allen School of Computer Science and Engineering, University of Washington
- **Jeff Dean/Heidi Hopper Endowed Regental Fellowship in Computer Science & Engineering** Sept. 2023
– Paul G. Allen School of Computer Science and Engineering, University of Washington
- **Dean's List Scholar** Jun. 2021, Jun. 2022, Jun. 2023
– Faculty of Arts and Science, University of Toronto
- **Dr. James A. & Connie P. Dickson Scholarship In Science & Mathematics** Sept. 2022, Sept. 2023
– “Given to the best students enrolled in science and mathematics programs.”
– University College, University of Toronto
- **Department of Computer Science Undergraduate Research Award** May 2022
– Department of Computer Science, University of Toronto
- **Galois Awards in Mathematics** Oct. 2021
– “Given to the best students enrolled in a mathematics specialist program.”
– University College, University of Toronto

SKILLS

- **Programming languages:** C/C++, Python, Java, Racket, and Haskell.
- **Assembly programming:** MIPS, ARM Aarch64, Intel x86 instruction sets.
- **Database management systems:** Microsoft Access, MySQL.
- **Mathematical computation and data analysis:** R, Mathematica, MATLAB.

LANGUAGES

- **English:** proficient
 - **IELTS:** Overall 8.0 (Aug. 2019)
- **Chinese:** native speaker